

Yajie Zhou

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EDUCATION

Nanchang University

Master of Medicine (M.Med.), Surgery | Average: 84.27/100

Nanchang, China

Sep 2024–Expected Jun 2027

Jinggangshan University

Bachelor of Medicine (B.Med.), Clinical Medicine | Average: 82.38/100

Ji'an, China

Sep 2019–Jun 2024

RESEARCH PROFILE

Clinically trained researcher in computational cancer biology investigating how tumor cell-state plasticity and metabolic reprogramming relate to the tumor microenvironment. Integrates bulk, single-cell, and spatial transcriptomics with statistical learning and targeted experimental follow-up to generate biologically interpretable, externally reproducible findings in cancer.

PUBLICATIONS

Peer-Reviewed Articles

- **Zhou, Y.**, Hu, Z., Xie, L., Zhang, W., & Rao, H. (2026). Comprehensive multi-omics profiling of tertiary lymphoid structures reveals immunogenetic landscapes and prognostic subtypes in lung adenocarcinoma. *Molecular Carcinogenesis*, 65(7), 888–903. <https://doi.org/10.1002/mc.70120>.
- **Zhou, Y.**, Hu, Z., Xie, L., Zhang, S., Huang, Q., Zhang, N., Liu, Y., Zhang, W., & Wang, K. (2026). Identification and external validation of a prognostic signature based on bone morphogenetic protein-related mRNAs for kidney renal clear cell carcinoma. *Discover Oncology*, 17(1), 1146. <https://doi.org/10.1007/s12672-026-05238-z>.
- Xie, L., **Zhou, Y.**, Hu, Z., Zhang, W., & Zhang, X. (2025). Integrative multi-omics reveals energy metabolism-related prognostic signatures and immunogenetic landscapes in lung adenocarcinoma. *Frontiers in Immunology*, 16, 1679464. <https://doi.org/10.3389/fimmu.2025.1679464>.

Manuscript under Review

- **Zhou, Y.**, Hu, Z., Xie, L., Lai, Y., Kuang, J., Zhang, W., & Xie, L. Multi-omics characterization of ENO1-associated mitophagy–metabolism crosstalk and tumor microenvironment remodeling in lung adenocarcinoma.

RESEARCH EXPERIENCE

Department of Thoracic Surgery, The Second Affiliated Hospital of Nanchang University

Nanchang, China

Graduate Researcher; Advisor: Wenxiong Zhang, MD, Professor of Thoracic Surgery

2024–Present

ENO1-Associated Mitophagy–Metabolism Crosstalk and Tumor Microenvironment Remodeling in Lung Adenocarcinoma

2025–Present

Project Lead and First Author; manuscript under review

- Investigated associations among ENO1-linked metabolic programs, malignant epithelial states, and spatial tumor microenvironment organization in lung adenocarcinoma through integrative multi-omics analysis.
- Integrated 11 LUAD single-cell samples (over 33,000 cells) with epithelial subclustering, Monocle3 pseudotime, RCTD spatial deconvolution, and CellChat; identified an ENO1-high malignant epithelial state enriched for glycolysis, MYC targets, and hypoxia within an epithelial–stromal niche.
- Developed and externally evaluated an ensemble model combining gradient boosting machine and random survival forest, achieving 5-year AUCs of 0.763 and 0.789 in two validation cohorts and prioritizing ENO1 for downstream investigation.
- Confirmed ENO1 expression patterns by RT-qPCR in HBE and three LUAD cell lines; led study design, computational analysis, biological interpretation, figure development, and manuscript drafting.

TLS-Related Immunogenetic Landscapes and Prognostic Subtypes in Lung Adenocarcinoma

2025–2026

Project Lead and First Author; published in Molecular Carcinogenesis

- Integrated TCGA-LUAD with eQTLGen data from 31,684 participants and a LUAD GWAS of 11,273 cases and 55,483 controls to identify genetically supported TLS-related candidates using five-method Mendelian randomization and transcriptomic filtering.
- Benchmarked 10 survival-learning algorithms across 117 configurations, with feature selection performed within leave-one-out cross-validation and validation in three independent GEO cohorts; selected a stepwise Cox + GBM model with TCGA 1-, 3-, and 5-year AUCs of 0.774, 0.762, and 0.789.
- Analyzed nine LUAD single-cell samples comprising 118,602 cells, localizing the 10-gene signature primarily to T/NK and myeloid compartments; confirmed differential expression in HBE and three LUAD cell lines by RT-qPCR.
- Led study design, computational analysis, experimental follow-up, biological interpretation, figure preparation, and manuscript writing.

BMP-Related mRNA Prognostic Modeling in Kidney Renal Clear Cell Carcinoma

2024–2026

Project Lead and First Author; published in Discover Oncology

- Developed a nine-gene BMP-related prognostic signature using 526 TCGA-KIRC cases and independently validated it in 101 patients from E-MTAB-1980.
- Applied differential-expression analysis, LASSO, and Cox modeling; achieved 5-year AUCs of 0.801, 0.720, and 0.763 in the training, test, and external-validation cohorts, respectively, with an integrated nomogram C-index of 0.775.
- Characterized risk-associated immune, genomic, and predicted therapeutic-response patterns, including higher tumor mutation burden and B-cell/immunoglobulin-related programs in the high-risk group.
- Confirmed differential expression of all nine signature genes by RT-qPCR across HK-2 and three KIRC cell lines; led data analysis, biological interpretation, figure preparation, and manuscript drafting.

TECHNICAL SKILLS

- **Single-Cell, Spatial, and Bulk Transcriptomics:** Seurat, Harmony, Monocle3, RCTD, CellChat, WGCNA, GSVA, clusterProfiler, and TCGAbiolinks; malignant epithelial-state characterization, spatial deconvolution, pathway analysis, and multi-cohort integration.
- **Experimental Validation:** RNA and protein extraction, reverse transcription, RT-qPCR, Western blotting, and quantitative gene/protein expression analysis.
- **Statistical Learning and Genetic Epidemiology:** Cox regression, LASSO-Cox, stepwise Cox, random survival forest, CoxBoost, GBM, survival-SVM, model benchmarking, LOOCV, external validation, and two-sample Mendelian randomization.
- **Programming and Reproducibility:** R; Git/GitHub; Markdown; LaTeX; version-controlled scripts, fixed random seeds, and documented analysis workflows.

RESEARCH MENTORING & COLLABORATION

Mentored undergraduate students in computational cancer research, providing guidance on analysis planning, R-based workflows, data visualization, and biological interpretation. Supported laboratory members with transcriptomic data processing, statistical analysis, troubleshooting, and figure preparation.

CLINICAL TRAINING

Completed medical coursework and multi-department clinical rotations in internal medicine and surgery, providing a translational perspective for cancer research and interpretation of multi-omics findings.

LANGUAGES

Mandarin Chinese (native); English (professional working proficiency).